SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: SP44 PTA Powder
Form: Powder
Size: 80/230, 100/270

PTP No 9
Rev: 2
Date: 5/12/2025

Specifications

AWS A5.21, Class ERNiCr-A (Chemistry Only)

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
All	38-45	12-18s/50g	3.8-5.0 g/cm^3

Chemistry (% by weight)						
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**B**
All	0.2-0.4 (aim 0.25)	2.8-3.3 (aim 3.2)	7.0-8.2 (aim 7.5)	BAL	2.0-3.0 (aim 2.5)	1.5-2.0 (aim 1.8)

Sieve Analysis (% by weight)

80/230	**+80**	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**Pan**
	0.0 max	BAL						5.0 max

100/270	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**+270**	**Pan**
	3.0 max	BAL						5.0 max

Approval

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: Fe27Co28 PTA Powder
Form: Powder
Size: 100/270

PTP No 37
Rev: 0
Date: 7/8/2025

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
100/270	**Hardness**	**Hall Flow**	**Apparent Density**
	N/R	12-18s/50g	4.5-6.0 g/cm^3

Chemistry (% by weight)						
Button Specimen	**C**	**Mn**	**Si**	**Ni**	**Fe**	**Co**
100/270	0.05 max	0.1-0.3	0.4-0.6	BAL	25-29	27-29

Sieve Analysis (% by weight)									
100/270	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**+270**	**Pan**
	2.0 max	BAL					10.0 max	5.0 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: FB-100G2 Spray Fuse Powder
Form: Powder
Size: 100/270

PTP No 51
Rev: 1
Date: 2/25/2026

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Spray Test	SPRAY-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
100/270	25-40	17s/50g max	4.0-5.0 g/cm^3

Chemistry (% by weight)						
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**B**
100/270	0.07 max	2.5-3.5	0.20 max	BAL	0.80 max	1.8-2.4

Sieve Analysis (% by weight)

100/270	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**+270**	**Pan**
	3 max	BAL						2 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: 431 Stainless PTA Powder
Form: Powder
Size: 100/270

PTP No 86
Rev: 0
Date: 04/08/2026

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
100/270	Report	Report	Report

Chemistry (% by weight)						
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**S**
100/270	0.15-0.25	1.0 max	15.0-17.0	1.25-2.50	BAL	0.03 max

Sieve Analysis (% by weight)							
100/270	**+120**	**+140**	**+170**	**+200**	**+230**	**+270**	**Pan**
	1 max	BAL					3 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: 316L Stainless PTA Powder
Form: Powder
Size: 100/270

PTP No 67
Rev: 0
Date: 12/09/2025

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
100/270	N/R	16-18s/50g	4.0-5.0 g/cm^3

Chemistry (% by weight)								
Button Specimen	**C**	**Mn**	**Si**	**Cr**	**Ni**	**Mo**	**Fe**	**Cu**
100/270	0.03 max	1.0-2.5	1.0 max	18.0-20.0	11.0-14.0	2.0-3.0	BAL	0.75 max

Sieve Analysis (% by weight)									
100/270	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**+270**	**Pan**
	0.0 max	BAL					15.0 max	3.0 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: 316 Laser Cladding Powder
Form: Powder
Size: 100/270

PTP No 18
Rev: 3 (Revised C to standard grade, remove AWS spec.)
Date: 5/06/2025

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
100/270	N/R	16-18s/50g	4.0-5.0 g/cm^3

Chemistry (% by weight)								
Button Specimen	**C**	**Mn**	**Si**	**Cr**	**Ni**	**Mo**	**Fe**	**Cu**
100/270	0.08 max	1.0-2.5	1.0 max	18.0-20.0	11.0-14.0	2.0-3.0	BAL	0.75 max

Sieve Analysis (% by weight)

100/270	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**+270**	**Pan**
	0.0 max	BAL					15.0 max	3.0 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: #6L Laser Cladding Powder
Form: Powder
Size: 100/270

PTP No 44
Rev: 0
Date: 7/18/2025

Specifications

AWS A 5.21, Class ERCoCr-A (Chemistry Only)

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
100/270	38-46	13-16s/50g	4.5-5.0 g/cm^3

Chemistry (% by weight)												
Button Specimen	**C**	**Mn**	**Si**	**Cr**	**Ni**	**Mo**	**W**	**Fe**	**Co**	**P**	**B**	**TAO**
100/270	0.9-1.2	0.5 max	0.8-1.5	26-28	1.5-3.0	0.5 max	3.0-5.0	1.5-3.0	BAL	0.03 max	0.01 max	0.50 max

Sieve Analysis (% by weight)

100/270	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**+270**	**Pan**
	0 max	BAL						3 max

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SURFACE ENGINEERING
Wear Resistance Specialists

Cobalt #21 Data Sheet

Description: Cobalt alloy #21 is a Molybdenum strengthened, high temperature, corrosion resistant alloy with exceptional impact, erosion , and metal to metal wear resistance. The alloy work hardens in service. It has moderate abrasion resistance as compared to Seaco Cobalt alloy #6.

Applications: Hot Forming Dies, Forging Dies, Erosion Shields, Valve Seats and plugs, Hydro-Turbine Runners, Pump shafts and sleeves.

Typical Chemical Analysis:

C	Mn	Si	Ni	Cr	W	Mo	Fe	Co
.25	.50	.60	2.5	26	-	5.5	1.0	Bal.

Machineability: Good

Hardness: Rc 22-32 as deposited, Rc 38-40 wok hardened

Hot Hardness (DPH): 800°F - 130; 1000°F - 135; 1200°F - 140; 1400°F - 110

Adhesive Wear Data (Avg.): 90lb Load - 2.5mm³; 150lb Load - 5.2mm³ 210lb Load - 10.3mm³; 300lb Load - 14.5mm³
(Volume Loss)

Abrasive Wear Data: Volume Loss of 86.4mm³ on ASTM G65B Test

Charpy Impact Strength (Un-Notched, Room Temperature): 27 Ft. Lb

Density: .299 Lbs/In³

Liquidus: 2480°F

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: SP50G2 PTA Powder
Form: Powder
Size: 100/230

PTP No 45
Rev: 0
Date: 7/22/2025

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
All	47-53 HRC	12-18s/50g	4-5 g/cm^3

Chemistry (% by weight)							
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**B**	**O2**
All	0.4 max	3.0-4.5	0.5 max	BAL	0.4 max	2.5-3.5	0.5 max

Sieve Analysis (% by weight)							
100/230	**+100**	**+120**	**+140**	**+170**	**+200**	**+230**	**Pan**
	3.0 max	BAL					3.0 max

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